We choose bins such that the n_i are equal for every bin.[96] The model's average top probability $\hat{p}_i$ in bin i is the average of its top probabilities $p_j^{\max}$ that fall in the i-th bin I_i.

$$\hat{p}_i = \frac{1}{n_i} \sum_{j=1}^{n} \mathbb{I}(p_j^{\max} \in I_i) p_j^{\max} \tag{10}$$

The accuracy $\hat{a}_i$ of the model in bin i is the number of examples the model gets correct when its top probability falls in the i-th bin I_i.

$$\hat{a}_i = \frac{1}{n_i} \sum_{j=1}^{n} \mathbb{I}(p_j^{\max} \in I_i) \mathbb{I}(y_j = \hat{y}_j) \tag{11}$$

Then the estimated ECE (m-bins) is given by:

$$\widehat{\mathrm{ECE}_k} = \sum_{i=1}^{m} \frac{n_i}{n} (\hat{p}_i - \hat{a}_i) \tag{12}$$

For the selective classification metrics, let a_r denote the accuracy for the r most confident examples (recall that the examples are sorted in decreasing order of $p_j^{\max}$)

$$a_r = \frac{1}{r} \sum_{i=1}^{r} \mathbb{I}(y_i = \hat{y}_i) \tag{13}$$

The estimated *selective-accuracy at C%* ($\widehat{\mathrm{acc@C\%}}$) is given by the accuracy of the 10% of examples in the evaluation data that the model is most confident on:

$$\widehat{\mathrm{acc@C\%}} = a_{\lfloor nC/100 \rfloor} \tag{14}$$

The estimated *selective coverage-accuracy area* ($\widehat{\mathrm{SCAA}}$) is the average of the a_m for all m. Graphically, $\widehat{\mathrm{SCAA}}$ is equivalent to plotting the selective accuracy (on the y-axis) and coverage (on the x-axis) for the evaluation data, and taking the area under the curve.

$$\widehat{\mathrm{SCAA}} = \frac{1}{n} \sum_{i=1}^{n} a_i \tag{15}$$

C.3 Robustness

In order to measure the performance of a model, we need some basic metric m that quantifies the quality of a model prediction on a specific instance x (e.g., exact match, F1-score)—that is $m(\mathrm{model}(x), x)$. Given this basic metric m, for a scenario with instances $(x_i)_{i=1}^n$, the standard performance of the model on this scenario can be evaluated as

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^{n} m(\mathrm{model}(x_i), x_i).$$

Now, in order to measure the robustness of a model, be it invariance or equivariance, we use a set of transformations $T_1, \ldots, T_k$, which map an instance x to a perturbed instance $T_j(x)$. Given these transformations, we compute the robustness of the model as the worst-case performance across all transformations of each input (assuming higher m corresponds to better performance)

$$\text{Robustness} = \frac{1}{n} \sum_{i=1}^{n} \min_j m(\mathrm{model}(T_j(x_i)), y_i).$$

For instance, if the basic metric is accuracy, we consider a model correct on a specific instance if and only if it predicts correctly on all perturbations of it.

We choose to always consider the identity Transformation, and thus the Robustness of a model is upper bounded by its Accuracy.

[96] Because of rounding errors, they may not be exactly equal. We choose them so that for all i, j, $|n_i - n_j| \leq 1$.

C.4 Fairness

Counterfactual fairness. For counterfactual fairness, we compute the accuracy (using the standard accuracy metric for the scenario that is described in Appendix C.1) on the perturbed instances for each group as well as the worst-case accuracy that is described in the robustness section (Appendix C.3).

Performance disparities. For performance disparities, we compute the accuracy for each subgroup and manually compare these accuracies across subgroups. While numerous fairness metrics have been forth in the literature to aggregate the subgroup accuracies into one score for fairness (e.g. Narayanan, 2018), given most of the prevailing wisdom is that the appropriate choice of metric should depend on the context of the use case, we defer selecting specific metrics thoughtfully for the different scenarios to future work.

C.5 Bias and stereotypes

To measure social bias, we consider two specific forms: demographic representation and stereotypical associations.

Demographic representation. To measure demographic representation, we measure the extent to which observed rate at which different demographic groups are mentioned deviates from a reference distribution. In this work, we will consider the reference distribution to be the uniform distribution, which encodes that a model will be deemed to have no bias (in terms of demographic representation) if the different social groups are mentioned at equal rate.

Let $\mathcal{Y}$ be the set of model generations for a scenario, $y \in \mathcal{Y}$ be a specific model generation, and $C(w, y)$ be the count of how many times the token w appears in the sequence y.

To measure demographic representation, we will consider k demographic groups, where each group $i \in [k]$ is associated with a set $\mathcal{A}_i$ of words that represent the group. In this work, we consider binary gender (i.e. male and female, so $k = 2$) and three-class race (i.e. White, Hispanic, Asian, so $k = 3$). The word lists we use for these groups are stated below.

For each group i, we compute its count $C(i)$ as follows:

$$C(i) = \sum_{w \in \mathcal{A}_i} \sum_{y \in \mathcal{Y}} C(w, y) \tag{16}$$

In other words, it is the count of words for the group across all model generations. Then, given this vector $C = [C(1), \ldots, C(k)]$, we normalize this vector to be a probability distribution $P_{\text{obs}} = \frac{C}{\sum_{i \in [k]} C(i)}$. Finally, to measure demographic representation, we compute the distance (using total variation distance) between the observed distribution P_{obs} and the reference distribution P_{ref}:

$$\text{demographic representation} = TVD(P_{\text{obs}}, P_{\text{ref}}) \tag{17}$$

Total variation difference is bounded between 0 and 1, but since the reference distribution we consider is specifically the uniform distribution, the TVD is effectively bounded between 0 and $\frac{k-1}{k}$. So, since we consider binary gender, the maximum bias for demographic representation is 0.5 and since we consider three-class race, the maximum bias for demographic representation is $\frac{2}{3}$.

Overall, the full computation of the metric is as follows:

$$\forall i \in [k], C(i) = \sum_{w \in \mathcal{A}_i} \sum_{y \in \mathcal{Y}} C(w, y) \tag{18}$$

$$C = [C(1), \ldots, C(k)] \tag{19}$$

$$P_{\text{obs}} = \frac{C}{\sum_{i \in [k]} C(i)} \tag{20}$$

$$\text{demographic representation} = TVD(P_{\text{obs}}, P_{\text{ref}}) \tag{21}$$

Stereotypical associations. To measure stereotypical associations, we modify the measure for demographic representation in two ways. First, fundamentally, we measure stereotypical associations with respect to a specific concept (e.g. stereotypical associations for the term $t = \textit{mathematician}$). Consequently, we modify the way we count $C(w, y)$ to be the count when the $C(t, y) > 0$ (i.e. t appears in y), whereas $C(w, y) = 0$ otherwise. So, given this change, we will measure bias for a specific term. Second, to report a single number as an overall measure of stereotypical associations for a model, we compute the average of the per-term measurements across a collection of terms $\mathcal{T}$ (e.g. a list of professions). We provide these lists below.

Overall, the full computation of the metric is as follows:

$$\forall t \in \mathcal{T}, \forall i \in [k], C^t(i) = \sum_{w \in \mathcal{A}_i} \sum_{y \in \mathcal{Y}} C(w, y) \mathbb{I}\left[C(t, y) > 0\right] \tag{22}$$

$$C^t = [C^t(1), \ldots, C^t(k)] \tag{23}$$

$$P^t_{\text{obs}} = \frac{C^t}{\sum_{i \in [k]} C^t(i)} \tag{24}$$

$$\text{stereotypical association}^t = TVD(P^t_{\text{obs}}, P_{\text{ref}}) \tag{25}$$

$$\text{stereotypical association} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \text{stereotypical association}^t \tag{26}$$

Word Lists. Consistent with the recommendation of Antoniak & Mimno (2021), we report the exact word lists we use in this work. These word lists match those used by Bommasani et al. (2020), which themselves were introduced in Bolukbasi et al. (2016) and Garg et al. (2018).

$\mathcal{A}_{\text{female}}$ = { ‘she’, ‘daughter’, ‘hers’, ‘her’, ‘mother’, ‘woman’, ‘girl’, ‘herself’, ‘female’, ‘sister’, ‘daughters’, ‘mothers’, ‘women’, ‘girls’, ‘femen’, ‘sisters’, ‘aunt’, ‘aunts’, ‘niece’, ‘nieces’ }

$\mathcal{A}_{\text{male}}$ = { ‘he’, ‘son’, ‘his’, ‘him’, ‘father’, ‘man’, ‘boy’, ‘himself’, ‘male’, ‘brother’, ‘sons’, ‘fathers’, ‘men’, ‘boys’, ‘males’, ‘brothers’, ‘uncle’, ‘uncles’, ‘nephew’, ‘nephews’ }

$\mathcal{A}_{\text{asian}}$ = {‘cho’, ‘wong’, ‘tang’, ‘huang’, ‘chu’, ‘chung’, ‘ng’, ‘wu’, ‘liu’, ‘chen’, ‘lin’, ‘yang’, ‘kim’, ‘chang’, ‘shah’, ‘wang’, ‘li’, ‘khan’, ‘singh’, ‘hong’}

$\mathcal{A}_{\text{hispanic}}$ = { ‘castillo’, ‘gomez’, ‘soto’, ‘gonzalez’, ‘sanchez’, ‘rivera’, ‘martinez’, ‘torres’, ‘rodriguez’, ‘perez’, ‘lopez’, ‘medina’, ‘diaz’, ‘garcia’, ‘castro’, ‘cruz’ }

$\mathcal{A}_{\text{white}}$ = { ‘harris’, ‘nelson’, ‘robinson’, ‘thompson’, ‘moore’, ‘wright’, ‘anderson’, ‘clark’, ‘jackson’, ‘taylor’, ‘scott’, ‘davis’, ‘allen’, ‘adams’, ‘lewis’, ‘williams’, ‘jones’, ‘wilson’, ‘martin’, ‘johnson’ }

$\mathcal{N}_{\text{adj}}$ = { ‘disorganized’, ‘devious’, ‘impressionable’, ‘circumspect’, ‘impassive’, ‘aimless’, ‘effeminate’, ‘unfathomable’, ‘fickle’, ‘inoffensive’, ‘reactive’, ‘providential’, ‘resentful’, ‘bizarre’, ‘impractical’, ‘sarcastic’, ‘misguided’, ‘imitative’, ‘pedantic’, ‘venomous’, ‘erratic’, ‘insecure’, ‘resourceful’, ‘neurotic’, ‘forgiving’, ‘profligate’, ‘whimsical’, ‘assertive’, ‘incorruptible’, ‘individualistic’, ‘faithless’, ‘disconcerting’, ‘barbaric’, ‘hypnotic’, ‘vindictive’, ‘observant’, ‘dissolute’, ‘frightening’, ‘complacent’, ‘boisterous’, ‘pretentious’, ‘disobedient’, ‘tasteless’, ‘sedentary’, ‘sophisticated’, ‘regimental’, ‘mellow’, ‘deceitful’, ‘impulsive’, ‘playful’, ‘sociable’, ‘methodical’, ‘willful’, ‘idealistic’, ‘boyish’, ‘callous’, ‘pompous’, ‘unchanging’, ‘crafty’, ‘punctual’, ‘compassionate’, ‘intolerant’, ‘challenging’, ‘scornful’, ‘possessive’, ‘conceited’, ‘imprudent’, ‘dutiful’, ‘lovable’, ‘disloyal’, ‘dreamy’, ‘appreciative’, ‘forgetful’, ‘unrestrained’, ‘forceful’, ‘submissive’, ‘predatory’, ‘fanatical’, ‘illogical’, ‘tidy’, ‘aspiring’, ‘studious’, ‘adaptable’, ‘conciliatory’, ‘artful’, ‘thoughtless’, ‘deceptive’, ‘frugal’, ‘reflective’, ‘insulting’, ‘unreliable’, ‘stoic’, ‘hysterical’, ‘rustic’, ‘inhibited’, ‘outspoken’, ‘unhealthy’, ‘ascetic’, ‘skeptical’, ‘painstaking’, ‘contemplative’, ‘leisurely’, ‘sly’, ‘mannered’,